Decays

Γ: decay width prob of decay / unit time.

$\tau = \frac{1}{\Gamma}$ lifetime

$\Gamma = \Gamma(i \to f)$

$a \longrightarrow b + c$

$a \longrightarrow b + c + d.$

Density of states

$$\Gamma = 2\pi \, |M_{if}|^2 \, \rho(E) \qquad \rho(E) = \left.\frac{dn}{dE}\right|_{E_f = E_i}$$

M_{if}: matrix element $M = \langle f | H_I | i \rangle$

Fermi's 2^{nd} Golden Rule

Relativistic Golden Rule

$1 \longrightarrow 2 + 3 + \cdots + n$

$|i\rangle = |1\rangle$

$|f\rangle = |2, \ldots n\rangle$

$M_{if} = \langle f | H_I | i \rangle$

conservation of En-mom.

$$\Gamma = S \frac{1}{2m_1} \int |M_{fi}|^2 (2\pi)^4 \, \delta^{(4)}(\underline{p}_1 - \underline{p}_2 - \underline{p}_3 - \cdots - \underline{p}_n) \times$$

$$\times \prod_{j=2}^{n} (2\pi) \, \delta(\underline{p}_j^2 - m_j^2) \, \theta(E_j) \frac{d^4 \underline{p}_j}{(2\pi)^4}$$

phase space for j-th particle

statistical factor

$\underline{p}_j$: 4-mom of j-th particle

$\underline{p}_j = (E_j, \vec{p}_j)$

$\underline{p}_j^2 = E_j^2 - \vec{p}_j^{\,2} = m_j^2$ for relativistic particles.

$\theta(E_j)$: $E_j > 0$

$a \to b + b + c + c + c$

$$S = \frac{1}{n_b!} \frac{1}{n_c!} = \frac{1}{2!} \frac{1}{3!} = \frac{1}{2} \frac{1}{6} = \frac{1}{12}$$

$K^0 \to \pi^0\pi^0\pi^0 \qquad S = \frac{1}{3!} = \frac{1}{6}$

$K^0 \to \pi^+\pi^- \qquad S = 1$

PDG: Particle Data Group

pdg.lbl.gov

$\pi^+ \to \mu^+\nu_\mu \qquad S = 1$

$H \to \gamma\gamma \qquad S = \frac{1}{2}$

Two-body Decay

$a \to b + c \qquad S = 1$

$$\Gamma = \frac{1}{2m_a}\int |M|^2 (2\pi)^4\, \delta^4(\underline{p}_a - \underline{p}_b - \underline{p}_c) \prod_{j=b,c} (2\pi)\,\delta(\underline{p}_j^2 - m_j^2)\,\theta(E_j)\, \frac{d^4 p_j}{(2\pi)^4}$$

$$\delta(\underline{p}_j^2 - m_j^2)\,\theta(E_j)$$

$$\hookrightarrow E_j^2 - (\vec{p}_j)^2 - m_j^2 = E_j^2 - (\vec{p}_j^2 + m_j^2) = f(x) = f(E)$$

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x)|_{x=x_i}} \qquad f(x_i) = \emptyset$$

$$f(E) = E^2 - (p^2 + M^2) \qquad E = \pm\sqrt{p^2 + m^2}$$

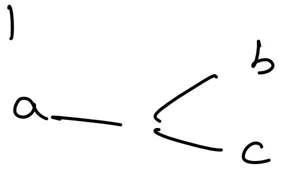
$$f'(E) = 2E$$

$$\delta(\underline{p}_j^2 - m_j^2)\,\theta(E_j)\,\frac{d^4 \underline{p}_j}{(2\pi)^4} = \frac{\delta(E - E_j)}{2\sqrt{\vec{p}_j^{\,2} + m_j^2}}\, \frac{d^4 \underline{p}_j}{(2\pi)^4}$$

$$d^4 \underline{p}_j = dE_j\, d^3\vec{p}_j$$

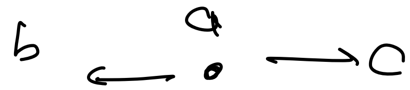
$$\Gamma_{a\to b+c} = \frac{1}{(2\pi)^2}\frac{1}{2m_a}\int |M|^2\, \delta^4(\underline{p}_a - \underline{p}_b - \underline{p}_c)\, \frac{d^3\vec{p}_b\, d^3\vec{p}_c}{4\sqrt{p_b^2 + m_b^2}\sqrt{p_c^2 + m_c^2}}$$

$$p_b \equiv |\vec{p}_b|$$



$$\delta^4(\underline{p}_a - \underline{p}_b - \underline{p}_c) = \delta(E_a - E_b - E_c)\,\delta^3(\vec{p}_a - \vec{p}_b - \vec{p}_c)$$

$$\vec{p}_b + \vec{p}_c = \vec{p}_a = \emptyset \implies \vec{p}_b = -\vec{p}_c$$



$|\vec{P}| = |\vec{P_b}| = |\vec{P_c}|$

$d^3\vec{p} = d\cos\theta \, d\varphi \, p^2 dp$

$d\Omega = d\cos\theta \, d\varphi$

$$\Gamma = \frac{1}{32\pi^2} \frac{1}{m_a^2} |\vec{P}| \int |M|^2 d\Omega$$

$|\vec{P}| = ?$

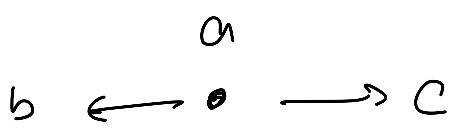
$m_a = E_b + E_c = \sqrt{m_b^2 + p^2} + \sqrt{m_c^2 + p^2}$

$|\vec{P}| = P = |\vec{P_b}| = |\vec{P_c}|$

$$P = \frac{m_a^2 - m_b^2 - m_c^2}{2m_a}$$

$$\Gamma \propto \frac{|\vec{P}|}{m_a^2} |M|^2$$

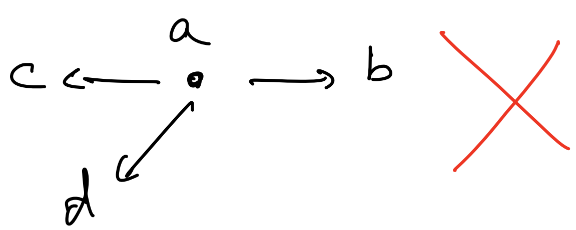
Nature of interaction



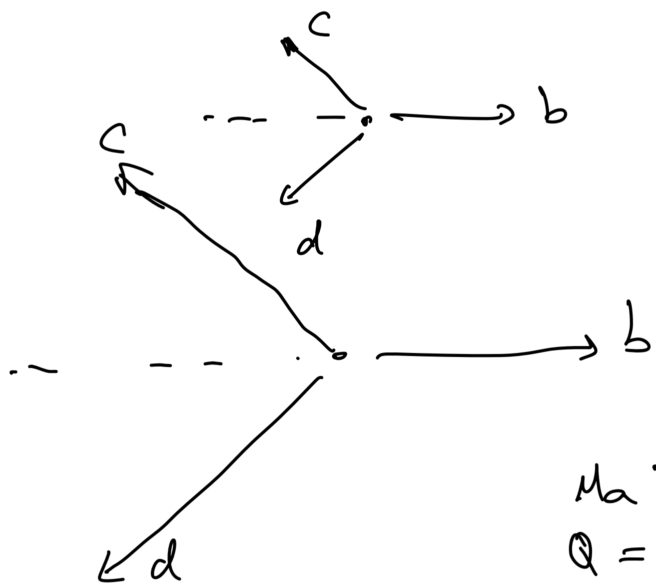
$Q = m_a - m_b - m_c$

3-body Decay

$a \to b + c + d$



No mom conserv.

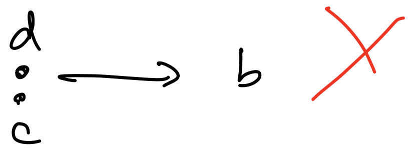


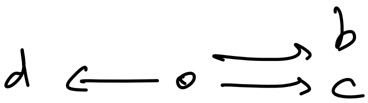
$\vec{P_b} + \vec{P_c} + \vec{P_d} = \phi$

$m_a > m_b > m_c > m_d$

$Q = m_a - m_b - m_c - m_d$

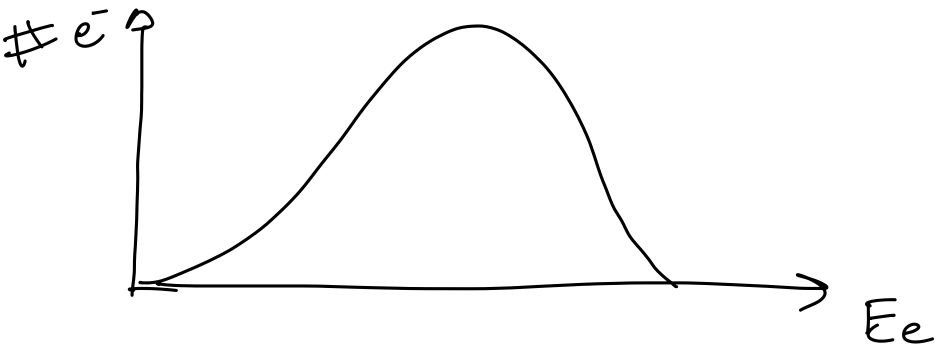
$\vec{P}_d = \phi$





P_b, P_c, P_d depend on M_a
m_b, m_c, m_d

$n \to p + e^- + \bar{\nu}_e$ 3- body spectrum.



$X \longrightarrow a + b$

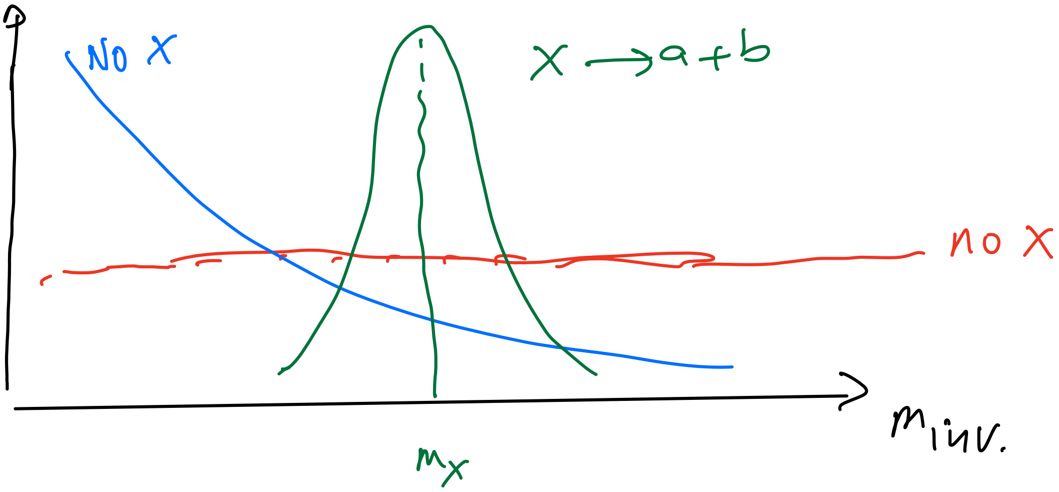
produce x in collision
observe a and b in your detector
measure $\vec{P}_a, E_a, \vec{P}_b, E_b$

$$(\underline{P_a} + \underline{P_b})^2 = (E_a + E_b)^2 - (\vec{P_a} + \vec{P_b})^2 = M_{ab}^2$$

invariant mass of a, b

$\underline{P_a} + \underline{P_b} \equiv \underline{P_x}$ conservation of $E, \vec{P}$

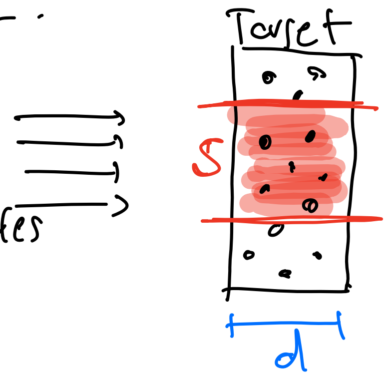
$(\underline{P_x})^2 = E_x^2 - \vec{P}_x^2 \equiv M_x^2$ invariant under Lorentz Transf.

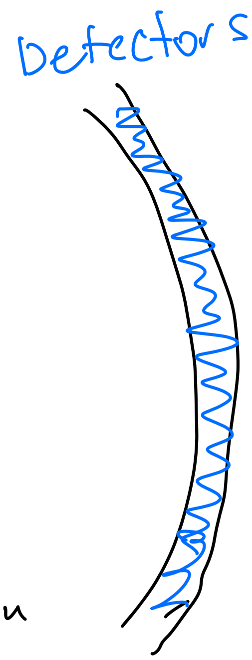


Collisions

$$a + b \longrightarrow c + d + \cdots$$

beam of particles





$$\frac{dN_r}{dt} = \sigma \, \frac{dN_p}{dt} \, n_b \cdot d$$

$$\frac{\#\text{interactions}}{\text{time interval}}$$

Volume of interaction

$V = S \cdot d$

N_B: # targets in V

$$n_b = \frac{N_B}{V}$$

$[n_b] = L^{-3}$

$[d] = L$

$[\sigma] = L^2$

Cross section for this process

$a + b \longrightarrow a + b$ elastic scattering.

$\alpha + N \longrightarrow \alpha + N$

$e^+ + e^- \longrightarrow e^+ + e^-$ elastic

$e^+ + e^- \longrightarrow \mu^+ + \mu^-$ inelastic scattering.

$p + p \longrightarrow p + p$

$p + p \longrightarrow p + p + p + \bar{p}, \quad p + p + \pi^+ + \pi^-$

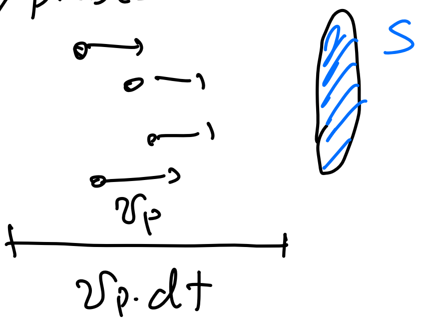
$X + N \longrightarrow X + e^+, e^-$. —

\# targets in interaction volume N_B

$$\frac{dN_r}{dt} = \sigma \frac{dN_p}{dt} \cdot n_b \cdot d = \sigma \underbrace{\frac{dN_p}{dt} \frac{1}{S}}_{\phi_p} \, S \cdot d \cdot n_b$$

ϕ_p : flux of incoming particles

incoming particles / probes



$$\phi_p = \frac{dN_p}{dt} \frac{1}{S} = n_p \cdot v_p$$

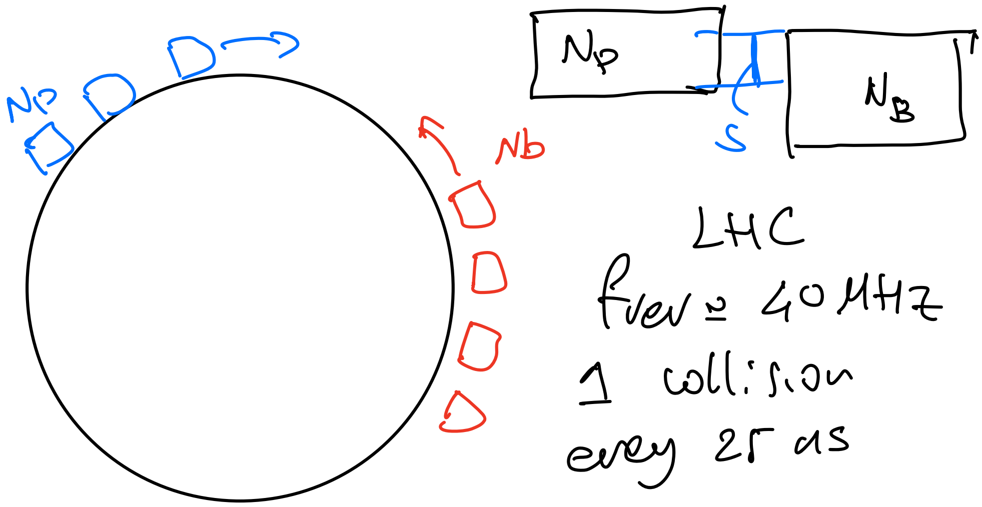
$$\frac{dN_r}{dt} = \sigma \cdot \phi_p \cdot N_b \Rightarrow \underbrace{\frac{1}{N_b} \frac{dN_r}{dt}}_{} = \sigma \cdot \phi_p$$

prob. of interaction per unit time per single target particle

$\sigma(a+b \to c+d)$: from Fermi's 2nd Golden Rule

In colliding beams: you control $\frac{dN_p}{dt}$

$\sim 10^{11}$ protons



$$\frac{dN_r}{dt} = \sigma N_b \phi_p$$

$$= \sigma N_b \frac{N_p}{S} f_{rev}$$

freq. of revolution.

LHC

$f_{rev} \simeq 40$ MHz

1 collision every 25 ns

$$\frac{dN_r}{dt} = \sigma \underbrace{\frac{N_b N_p}{S} f}_{\text{properties of the collider}} = \sigma \cdot \mathcal{L}$$

physics to study

$\mathcal{L}$: instantaneous luminosity of the collider

$[\sigma] = L^2 \qquad cm^2$

$$[\mathcal{L}] = \frac{1}{cm^2}\frac{1}{s} = cm^{-2}\,Hz$$

$1\ fm = 10^{-15}\ m = 10^{-13}\ cm$

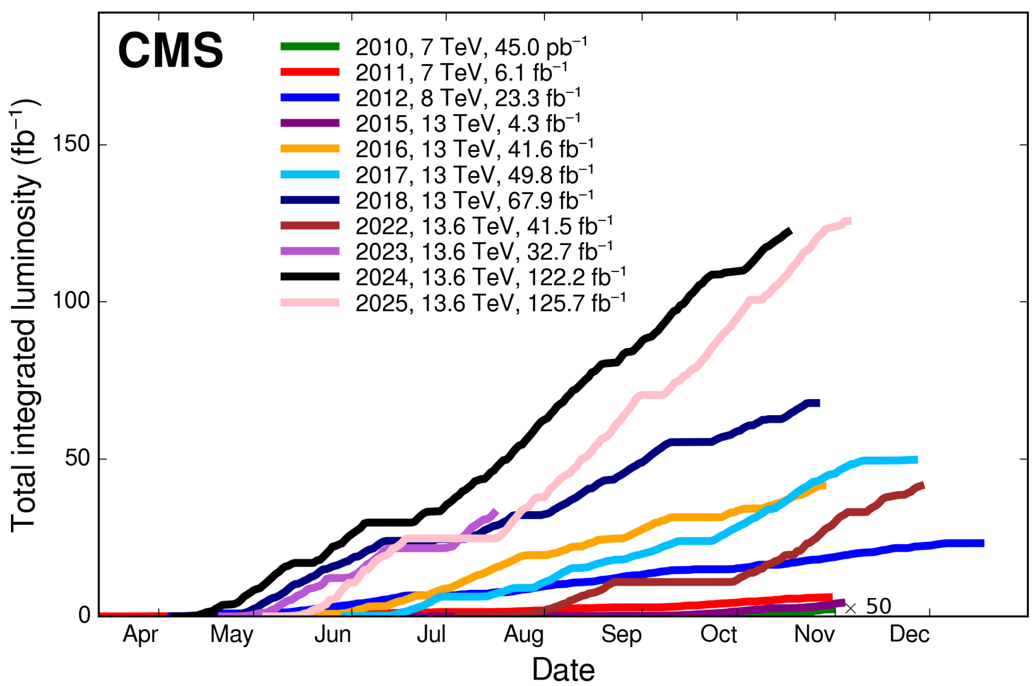
$1\ \text{barn} = 1b = 10^{-24}\ cm^2 = 10^{-28}\ m^2$

$[\mathcal{L}] = b^{-1} s^{-1}$

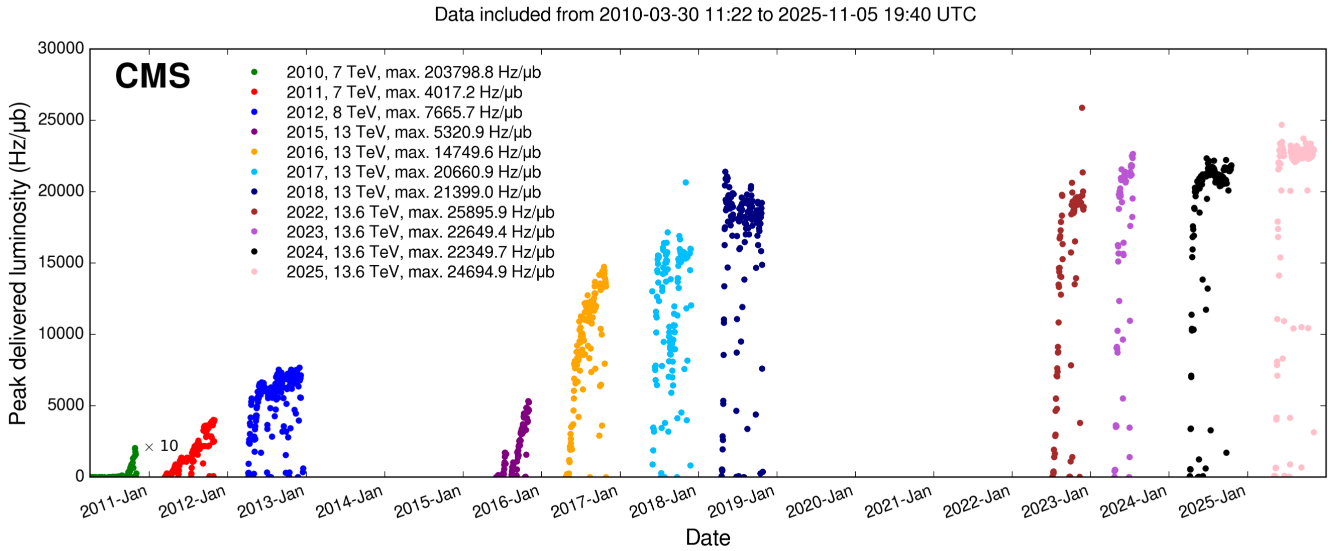
$$\#\ \text{events} = \frac{dN_r}{dt} \cdot \underbrace{\Delta t}_{\text{time of data taking.}} = \sigma \cdot \underbrace{\mathcal{L} \cdot \Delta t}_{= L_{int}}$$

integrated luminosity

$[L_{int}] = [\sigma]^{-1} = b^{-1}$



slope: instantaneous luminosity



Instantaneous lumi $\mathcal{L}$: # events / sec

Energy of the collider:

$$\underline{P}_1 \longrightarrow \qquad \longleftarrow \underline{P}_2$$

$$\sqrt{s} = \sqrt{(\underline{P}_1 + \underline{P}_2)^2}$$ Total energy available in the collision.

symmetric because: $\underline{P}_1 = (E, \vec{p})$

$\underline{P}_2 = (E, -\vec{p})$

$$\underline{P}_1 + \underline{P}_2 = (E + E, \vec{0})$$

$$\sqrt{s} = \sqrt{(2E)^2} = 2E$$

LHC: $E = 6.5$ TeV $= 6.5 \times 10^{12}$ eV

$$a + b \longrightarrow c + d$$

$$p + p \longrightarrow X + \bar{X} \qquad m_X \leq E$$